B.Tech. Degree II Semester Regular Examination July 2024

CE/ME/SE 23-200-0203A ENGINEERING MECHANICS

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Explain principles and theorems related to rigid body mechanics.
CO2: Identify the components of a system of forces acting on the rigid body.
CO3: Apply the conditions of equilibrium to various practical problems involving different force system.
CO4: Choose appropriate theorems, principles or formulae to solve problems of mechanics.
CO5: Solve problems involving rigid bodies, applying the properties of distributed areas and masses.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

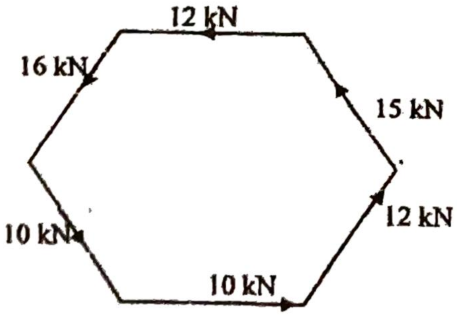
(Answer *ALL* questions)

		$(5 \times 2 = 10)$	Marks	BL	CO	PO
I.	(a)	List out and explain system of forces.	2	L1	2	1,2
	(b)	Define coefficient of friction. Show that the coefficient of friction is tangent of the angle of friction.	2	L1	3	1,2
	(c)	Determine the magnitude of two forces, such that if they act at right angle, their resultant is $\sqrt{10}$ N and if they act at 60°, their resultant is $\sqrt{13}$ N.	2	L3	2	1,2
	(d)	State D'Alembert's principle. Draw the free body diagram of a lift of weight 'W', moving upwards with an acceleration 'a' and also write the equations of dynamic equilibrium using this principle.	2	L1	4	1,2
	(e)	The curvilinear motion of a particle is defined by $v_x = 50 - 16t$ and $y = 100 - 4t^2$, where v_x is in meters per second, y is in meters and t is in seconds. It is also known that $x = 0$ when $t = 0$. Determine its velocity and acceleration when the position $y = 0$ is reached.	2	L3	5	1,2

PART B

$(4 \times 10 = 40)$

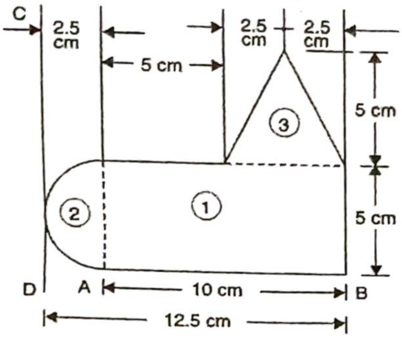
		Marks	BL	CO	PO
II.	Six forces of magnitude 10 kN, 12 kN, 15 kN, 12 kN, 16 kN and 10 kN are acting along the sides of the regular hexagon of side 2 m in order as shown in the figure. Determine the resultant force and its direction.	10	L3	2	1,2



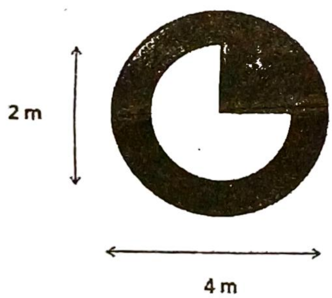
OR

(P.T.O.)

		Marks	BL	CO	PO
III.	Determine the centroid of the plane uniform lamina about an axis passing through the line CD shown in the figure.	10	L3	5	1,2

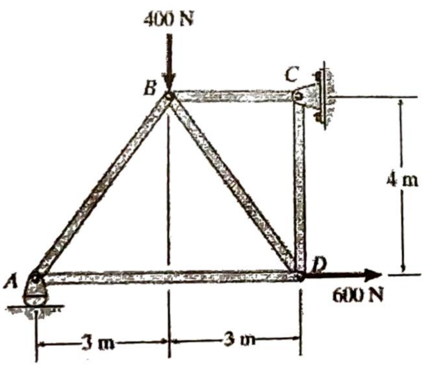


IV.	From the circular lamina of diameter 4 m, 3/4th quarter circle of diameter 2 m has been removed from the center. Determine the moment of inertia of the resulting composite section about the centroidal X axis.	10	L3	5	1,2



OR

V.	Determine the forces in each member the truss shown in the figure. Indicate whether the members are in tension or compression.	10	L3	3	1,2



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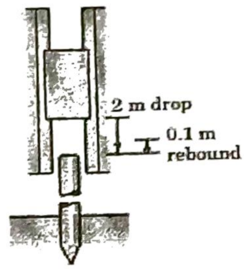
			Marks	BL	CO	PO
VI.		The ram of a pile driver has a mass of 800 kg and is released from rest 2 m above the top of the 2400 kg pile. If the ram rebounds to a height of 0.1 m after impact with the pile, calculate (i) the velocity of the pile immediately after impact (ii) the coefficient of restitution e (iii) the percentage loss of energy due to the impact.	10	L3	4	1,2

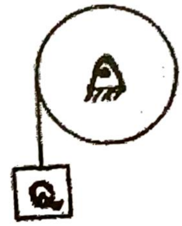


OR

			Marks	BL	CO	PO
VII.		Two bodies of weights 60 N and 40 N are connected to the two ends of a light inextensible string, which passes over a smooth pulley. The weight 60 N is placed on a smooth inclined plane of angle of inclination of 10°, while the weight 40 N is hanging free in air. Determine acceleration and tension in the string.	10	L3	4	1,2
VIII.	(a)	A ball is kicked with an initial velocity of 30 m/s at an angle of elevation 50°. Find its velocity after 2 seconds, at the highest point of its path and at a height of 6 m.	5	L1	5	1,2
	(b)	The armature of an electric motor has a speed of 1800 rpm, at the instant when the power cut off. If it comes to rest in 60 seconds, calculate the angular deceleration assuming it to be constant. How many complete revolutions does the armature make during this period?	5	L1	5	1,2

OR

			Marks	BL	CO	PO
IX.		A solid right circular drum of radius r = 30 cm and weight W = 150 N is free to rotate about its geometric axis. A flexible cord carrying a weight of Q = 50 N is wound around the circumference of the drum. If the weight Q is released from rest, find the time t required for it to fall through a height h = 3 m. With what velocity will it strike the floor.	10	L1	5	1,2



Bloom's Taxonomy Levels
L1 – 43 %, L3 – 57%.

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